

Progress: 500/2744 (18.2%)
Progress: 600/2744 (21.9%)
Progress: 700/2744 (25.5%)
Progress: 800/2744 (29.2%)
Progress: 900/2744 (32.8%)
Progress: 1000/2744 (36.4%)
Progress: 1100/2744 (40.1%)
Progress: 1200/2744 (43.7%)
Progress: 1300/2744 (47.4%)
Progress: 1400/2744 (51.0%)
Progress: 1500/2744 (54.7%)
Progress: 1600/2744 (58.3%)
Progress: 1700/2744 (62.0%)
Progress: 1800/2744 (65.6%)
Progress: 1900/2744 (69.2%)
Progress: 2000/2744 (72.9%)
Progress: 2100/2744 (76.5%)
Progress: 2200/2744 (80.2%)
Progress: 2300/2744 (83.8%)
Progress: 2400/2744 (87.5%)
Progress: 2500/2744 (91.1%)
Progress: 2600/2744 (94.8%)
Progress: 2700/2744 (98.4%)

GARCH Parameter Stability:

alpha: 0.1449 ± 0.0253

beta: 0.8264 ± 0.0319

alpha+beta: 0.9713

nu: 6.30 ± 1.58

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Processing: EUROSTOXX50

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Rolling estimation over 917 windows

Progress: 0/916 (0.0%)

Progress: 100/916 (10.9%)

Progress: 200/916 (21.8%)

Progress: 300/916 (32.8%)
Progress: 400/916 (43.7%)
Progress: 500/916 (54.6%)
Progress: 600/916 (65.5%)
Progress: 700/916 (76.4%)
Progress: 800/916 (87.3%)
Progress: 900/916 (98.3%)

GARCH Parameter Stability:

alpha: 0.1782 ± 0.0162
beta: 0.7940 ± 0.0239
alpha+beta: 0.9721
nu: 4.40 ± 0.32

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Processing: CAC40_FRANCE

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Rolling estimation over 1796 windows

Progress: 0/1795 (0.0%)
Progress: 100/1795 (5.6%)
Progress: 200/1795 (11.1%)
Progress: 300/1795 (16.7%)
Progress: 400/1795 (22.3%)
Progress: 500/1795 (27.9%)
Progress: 600/1795 (33.4%)
Progress: 700/1795 (39.0%)
Progress: 800/1795 (44.6%)
Progress: 900/1795 (50.1%)
Progress: 1000/1795 (55.7%)
Progress: 1100/1795 (61.3%)
Progress: 1200/1795 (66.9%)
Progress: 1300/1795 (72.4%)
Progress: 1400/1795 (78.0%)
Progress: 1500/1795 (83.6%)
Progress: 1600/1795 (89.1%)
Progress: 1700/1795 (94.7%)

GARCH Parameter Stability:

alpha: 0.1574 ± 0.0177

beta: 0.8212 ± 0.0277

alpha+beta: 0.9786

nu: 4.88 ± 0.62

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Processing: DAX_GERMANY

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Rolling estimation over 1898 windows

Progress: 0/1897 (0.0%)

Progress: 100/1897 (5.3%)

Progress: 200/1897 (10.5%)

Progress: 300/1897 (15.8%)

Progress: 400/1897 (21.1%)

Progress: 500/1897 (26.4%)

Progress: 600/1897 (31.6%)

Progress: 700/1897 (36.9%)

Progress: 800/1897 (42.2%)

Progress: 900/1897 (47.4%)

Progress: 1000/1897 (52.7%)

Progress: 1100/1897 (58.0%)

Progress: 1200/1897 (63.3%)

Progress: 1300/1897 (68.5%)

Progress: 1400/1897 (73.8%)

Progress: 1500/1897 (79.1%)

Progress: 1600/1897 (84.3%)

Progress: 1700/1897 (89.6%)

Progress: 1800/1897 (94.9%)

GARCH Parameter Stability:

alpha: 0.1296 ± 0.0317

beta: 0.8509 ± 0.0428

alpha+beta: 0.9805

nu: 4.95 ± 0.87

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Processing: AEX_NETHERLANDS

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Rolling estimation over 1672 windows

Progress: 0/1671 (0.0%)
Progress: 100/1671 (6.0%)
Progress: 200/1671 (12.0%)
Progress: 300/1671 (18.0%)
Progress: 400/1671 (23.9%)
Progress: 500/1671 (29.9%)
Progress: 600/1671 (35.9%)
Progress: 700/1671 (41.9%)
Progress: 800/1671 (47.9%)
Progress: 900/1671 (53.9%)
Progress: 1000/1671 (59.8%)
Progress: 1100/1671 (65.8%)
Progress: 1200/1671 (71.8%)
Progress: 1300/1671 (77.8%)
Progress: 1400/1671 (83.8%)
Progress: 1500/1671 (89.8%)
Progress: 1600/1671 (95.8%)

GARCH Parameter Stability:

alpha: 0.1532 ± 0.0177
beta: 0.8233 ± 0.0224
alpha+beta: 0.9765
nu: 5.50 ± 0.44

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Processing: HANG_SENG

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Rolling estimation over 1901 windows

Progress: 0/1900 (0.0%)
Progress: 100/1900 (5.3%)
Progress: 200/1900 (10.5%)
Progress: 300/1900 (15.8%)

Progress: 400/1900 (21.1%)
Progress: 500/1900 (26.3%)
Progress: 600/1900 (31.6%)
Progress: 700/1900 (36.8%)
Progress: 800/1900 (42.1%)
Progress: 900/1900 (47.4%)
Progress: 1000/1900 (52.6%)
Progress: 1100/1900 (57.9%)
Progress: 1200/1900 (63.2%)
Progress: 1300/1900 (68.4%)
Progress: 1400/1900 (73.7%)
Progress: 1500/1900 (78.9%)
Progress: 1600/1900 (84.2%)
Progress: 1700/1900 (89.5%)
Progress: 1800/1900 (94.7%)

GARCH Parameter Stability:

alpha: 0.0551 ± 0.0110

beta: 0.9253 ± 0.0259

alpha+beta: 0.9804

nu: 6.67 ± 0.70

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Processing: SENSEX

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Rolling estimation over 1380 windows

Progress: 0/1379 (0.0%)

Progress: 100/1379 (7.3%)

Progress: 200/1379 (14.5%)

Progress: 300/1379 (21.8%)

Progress: 400/1379 (29.0%)

Progress: 500/1379 (36.3%)

Progress: 600/1379 (43.5%)

Progress: 700/1379 (50.8%)

Progress: 800/1379 (58.0%)

Progress: 900/1379 (65.3%)

Progress: 1000/1379 (72.5%)

Progress: 1100/1379 (79.8%)

Progress: 1200/1379 (87.0%)

Progress: 1300/1379 (94.3%)

GARCH Parameter Stability:

alpha: 0.0899 ± 0.0112

beta: 0.8839 ± 0.0111

alpha+beta: 0.9739

nu: 6.81 ± 0.61

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Processing: RUSSELL2000

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Rolling estimation over 1906 windows

Progress: 0/1905 (0.0%)

Progress: 100/1905 (5.2%)

Progress: 200/1905 (10.5%)

Progress: 300/1905 (15.7%)

Progress: 400/1905 (21.0%)

Progress: 500/1905 (26.2%)

Progress: 600/1905 (31.5%)

Progress: 700/1905 (36.7%)

Progress: 800/1905 (42.0%)

Progress: 900/1905 (47.2%)

Progress: 1000/1905 (52.5%)

Progress: 1100/1905 (57.7%)

Progress: 1200/1905 (63.0%)

Progress: 1300/1905 (68.2%)

Progress: 1400/1905 (73.5%)

Progress: 1500/1905 (78.7%)

Progress: 1600/1905 (84.0%)

Progress: 1700/1905 (89.2%)

Progress: 1800/1905 (94.5%)

Progress: 1900/1905 (99.7%)

GARCH Parameter Stability:

alpha: 0.1152 ± 0.0246

beta: 0.8577 ± 0.0176

alpha+beta: 0.9729

nu: 9.61 ± 2.50

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Processing: SP500

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Rolling estimation over 4899 windows

Progress: 0/4898 (0.0%)

Progress: 100/4898 (2.0%)

Progress: 200/4898 (4.1%)

Progress: 300/4898 (6.1%)

Progress: 400/4898 (8.2%)

Progress: 500/4898 (10.2%)

Progress: 600/4898 (12.2%)

Progress: 700/4898 (14.3%)

Progress: 800/4898 (16.3%)

Progress: 900/4898 (18.4%)

Progress: 1000/4898 (20.4%)

Progress: 1100/4898 (22.5%)

Progress: 1200/4898 (24.5%)

Progress: 1300/4898 (26.5%)

Progress: 1400/4898 (28.6%)

Progress: 1500/4898 (30.6%)

Progress: 1600/4898 (32.7%)

Progress: 1700/4898 (34.7%)

Progress: 1800/4898 (36.7%)

Progress: 1900/4898 (38.8%)

Progress: 2000/4898 (40.8%)

Progress: 2100/4898 (42.9%)

Progress: 2200/4898 (44.9%)

Progress: 2300/4898 (47.0%)

Progress: 2400/4898 (49.0%)

Progress: 2500/4898 (51.0%)

Progress: 2600/4898 (53.1%)

Progress: 2700/4898 (55.1%)

Progress: 2800/4898 (57.2%)

Progress: 2900/4898 (59.2%)
Progress: 3000/4898 (61.2%)
Progress: 3100/4898 (63.3%)
Progress: 3200/4898 (65.3%)
Progress: 3300/4898 (67.4%)
Progress: 3400/4898 (69.4%)
Progress: 3500/4898 (71.5%)
Progress: 3600/4898 (73.5%)
Progress: 3700/4898 (75.5%)
Progress: 3800/4898 (77.6%)
Progress: 3900/4898 (79.6%)
Progress: 4000/4898 (81.7%)
Progress: 4100/4898 (83.7%)
Progress: 4200/4898 (85.7%)
Progress: 4300/4898 (87.8%)
Progress: 4400/4898 (89.8%)
Progress: 4500/4898 (91.9%)
Progress: 4600/4898 (93.9%)
Progress: 4700/4898 (96.0%)
Progress: 4800/4898 (98.0%)

GARCH Parameter Stability:

alpha: 0.1377 ± 0.0565

beta: 0.8418 ± 0.0673

alpha+beta: 0.9795

nu: 7.93 ± 6.18

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Processing: HANG_SENG_CHINA_ENT

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Rolling estimation over 1578 windows

Progress: 0/1577 (0.0%)

Progress: 100/1577 (6.3%)

Progress: 200/1577 (12.7%)

Progress: 300/1577 (19.0%)

Progress: 400/1577 (25.4%)

Progress: 500/1577 (31.7%)

Progress: 600/1577 (38.0%)
Progress: 700/1577 (44.4%)
Progress: 800/1577 (50.7%)
Progress: 900/1577 (57.1%)
Progress: 1000/1577 (63.4%)
Progress: 1100/1577 (69.8%)
Progress: 1200/1577 (76.1%)
Progress: 1300/1577 (82.4%)
Progress: 1400/1577 (88.8%)
Progress: 1500/1577 (95.1%)

GARCH Parameter Stability:

alpha: 0.0604 ± 0.0138
beta: 0.9180 ± 0.0215
alpha+beta: 0.9784
nu: 8.04 ± 0.66

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Processing: NIKKEI225

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Rolling estimation over 2975 windows

Progress: 0/2974 (0.0%)
Progress: 100/2974 (3.4%)
Progress: 200/2974 (6.7%)
Progress: 300/2974 (10.1%)
Progress: 400/2974 (13.4%)
Progress: 500/2974 (16.8%)
Progress: 600/2974 (20.2%)
Progress: 700/2974 (23.5%)
Progress: 800/2974 (26.9%)
Progress: 900/2974 (30.3%)
Progress: 1000/2974 (33.6%)
Progress: 1100/2974 (37.0%)
Progress: 1200/2974 (40.3%)
Progress: 1300/2974 (43.7%)
Progress: 1400/2974 (47.1%)
Progress: 1500/2974 (50.4%)

Progress: 1600/2974 (53.8%)
Progress: 1700/2974 (57.2%)
Progress: 1800/2974 (60.5%)
Progress: 1900/2974 (63.9%)
Progress: 2000/2974 (67.2%)
Progress: 2100/2974 (70.6%)
Progress: 2200/2974 (74.0%)
Progress: 2300/2974 (77.3%)
Progress: 2400/2974 (80.7%)
Progress: 2500/2974 (84.1%)
Progress: 2600/2974 (87.4%)
Progress: 2700/2974 (90.8%)
Progress: 2800/2974 (94.1%)
Progress: 2900/2974 (97.5%)

GARCH Parameter Stability:

alpha: 0.1293 ± 0.0333

beta: 0.8159 ± 0.0663

alpha+beta: 0.9453

nu: 6.90 ± 2.62

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Processing: NIFTY50

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Rolling estimation over 873 windows

Progress: 0/872 (0.0%)

Progress: 100/872 (11.5%)

Progress: 200/872 (22.9%)

Progress: 300/872 (34.4%)

Progress: 400/872 (45.9%)

Progress: 500/872 (57.3%)

Progress: 600/872 (68.8%)

Progress: 700/872 (80.3%)

Progress: 800/872 (91.7%)

GARCH Parameter Stability:

alpha: 0.0943 ± 0.0131

beta: 0.8769 ± 0.0143

alpha+beta: 0.9711

nu: 6.66 ± 0.72

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Processing: STI_SINGAPORE

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Rolling estimation over 1876 windows

Progress: 0/1875 (0.0%)

Progress: 100/1875 (5.3%)

Progress: 200/1875 (10.7%)

Progress: 300/1875 (16.0%)

Progress: 400/1875 (21.3%)

Progress: 500/1875 (26.7%)

Progress: 600/1875 (32.0%)

Progress: 700/1875 (37.3%)

Progress: 800/1875 (42.7%)

Progress: 900/1875 (48.0%)

Progress: 1000/1875 (53.3%)

Progress: 1100/1875 (58.7%)

Progress: 1200/1875 (64.0%)

Progress: 1300/1875 (69.3%)

Progress: 1400/1875 (74.7%)

Progress: 1500/1875 (80.0%)

Progress: 1600/1875 (85.3%)

Progress: 1700/1875 (90.7%)

Progress: 1800/1875 (96.0%)

GARCH Parameter Stability:

alpha: 0.1122 ± 0.0347

beta: 0.8201 ± 0.0946

alpha+beta: 0.9322

nu: 6.95 ± 0.97

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Processing: DJIA

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Rolling estimation over 1688 windows

Progress: 0/1687 (0.0%)
Progress: 100/1687 (5.9%)
Progress: 200/1687 (11.9%)
Progress: 300/1687 (17.8%)
Progress: 400/1687 (23.7%)
Progress: 500/1687 (29.6%)
Progress: 600/1687 (35.6%)
Progress: 700/1687 (41.5%)
Progress: 800/1687 (47.4%)
Progress: 900/1687 (53.3%)
Progress: 1000/1687 (59.3%)
Progress: 1100/1687 (65.2%)
Progress: 1200/1687 (71.1%)
Progress: 1300/1687 (77.1%)
Progress: 1400/1687 (83.0%)
Progress: 1500/1687 (88.9%)
Progress: 1600/1687 (94.8%)

GARCH Parameter Stability:

alpha: 0.1701 ± 0.0333
beta: 0.8088 ± 0.0210
alpha+beta: 0.9789
nu: 6.02 ± 0.89

Saved results: reports_garch/garch_all_models_summary.csv

model horizon

FIGARCH 1 5.012614

5 3.869530
10 3.729978
21 3.630639
63 3.511431
100 3.453819

GARCH 1 5.056123

5 3.904762
10 3.762132
21 3.657399

	63	3.532489
	100	3.469318
GJR	1	5.011928
	5	3.868457
	10	3.728432
	21	3.628112
	63	3.508384
	100	3.450308
IGARCH	1	4.134784
	5	2.964698
	10	2.879828
	21	2.845701
	63	2.791596
	100	2.794378

Name: qlike, dtype: float64